

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399724
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Takahashi; Junya et al.

Display control device for vehicle, vehicle display device, vehicle, method and non-transitory storage medium

Abstract

A display control device for a vehicle includes a memory, and at least processor. The at least one processor is coupled to the memory, is configured to execute a first OS, which carries out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands, and, in a case in which there is a problem with or a possibility of a problem with display of the display device, carry out display by only control of the second OS at the display device.

Inventors:	Takahashi; Junya (Tokyo-to, JP), Nishida; Masafumi (Kiyosu, JP)
Applicant:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Family ID:	1000008782512
Assignee:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Appl. No.:	18/058301
Filed:	November 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230221964 A1	Jul. 13, 2023

Foreign Application Priority Data

JP	2022-003398	Jan. 12, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: G06F9/44 (20180101); G06F3/14 (20060101); G06F9/445 (20180101)

U.S. Cl.:

CPC G06F9/44505 (20130101); G06F3/14 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2015/0205616	12/2014	Thai	345/542	G06F 9/4401
2018/0210774	12/2017	Young	N/A	G06F 11/0751
2022/0100451	12/2021	Ino	N/A	N/A
2022/0100574	12/2021	Nagano	N/A	N/A
2023/0102579	12/2022	Liu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111371875	12/2019	CN	N/A
113326070	12/2020	CN	N/A
2020201761	12/2019	JP	N/A
2020201762	12/2019	JP	N/A

Primary Examiner: Dorais; Craig C

Attorney, Agent or Firm: Dickinson Wright, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-003398 filed on Jan. 12, 2022, the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

(2) The present disclosure relates to a display control device for a vehicle, a vehicle display device, a vehicle, a method and a non-transitory storage medium.

Related Art

(3) Japanese Patent Application Laid-Open (JP-A) No. 2020-201761 for example discloses a control device for a vehicle that, at the time of start-up of an integrated ECU that carries out display control of a display within a vehicle cabin, carries out display control by, before start-up of the general-purpose OS, starting-up an RTOS (Real-Time Operating System) that generates preceding display elements that are to be displayed before start-up.

(4) However, in the device disclosed in JP-A No. 2020-201761, there is room for improvement in

devising display control in a case in which there is a problem or there is the possibility of a problem with display of the display device, after the display control at the general-purpose OS. In particular, in a vehicle display device, display for the user must always be ensured.

SUMMARY

(5) The present disclosure provides a display control device for a vehicle, a vehicle display device, a vehicle, a method and a non-transitory storage medium that ensure the display for a user in a case in which display control for a display device installed at a vehicle is carried out by plural OSs.

(6) A display control device for a vehicle of a first aspect includes a memory, and at least one processor connected to the memory, wherein the at least one processor is configured to execute a first OS, which carries out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands, and, in a case in which there is a problem with or a possibility of a problem with display of the display device, carry out display by only control of the second OS at the display device.

(7) In the display control device for a vehicle of the first aspect, display control of a display device installed at a vehicle is carried out by a processor executing a first OS and a second OS. The display device is configured to be able to be seen by a user.

(8) In a case in which there is a problem with or the possibility of a problem with displaying by the display device, the at least one processor carries out display on the display device by only control by the second OS. Therefore, in accordance with the present structure, display for the user can be ensured in cases in which display control of a display device installed at a vehicle is carried out by plural OSs.

(9) In a display control device for a vehicle of a second aspect, in the display control device for a vehicle of the first aspect, the at least one processor is configured be able to switch between display by control of the first OS and display by control of the second OS, and, in a case in which there is a problem with or a possibility of a problem with display of the display device, carries out display by control of the second OS instead of display by control of the first OS at the display device.

(10) In the display control device for a vehicle of the second aspect, in a case in which there is a problem with or the possibility of a problem with displaying by the display device by the first OS, display for the user can be ensured by switching to display by the second OS.

(11) In a display control device for a vehicle of a third aspect, in the display control device for a vehicle of the first aspect, the at least one processor is configured to be able to carry out superposed display of display by control of the first OS and display by control of the second OS at the display device, and, in a case in which there is a problem with or a possibility of a problem with display of the display device, stops display by control of the first OS at the display device.

(12) In the display control device for a vehicle of the third aspect, in a case in which there is a problem with or the possibility of a problem with superposed display at the display device by the first OS and the second OS, display for the user can be ensured by switching to display by the second OS only. Further, in a case of carrying out display control by plural OSs, due to there no longer being the need to carry out control for displaying similar information, an improvement in efficiency at the time of manufacturing the manufactured product, and a reduction in the memory capacity for information, and the like are possible.

(13) In a display control device for a vehicle of a fourth aspect, in the display control device for a vehicle of any one of the first aspect to the third aspect, in a case in which the problem disappears, the at least one processor is configured to carry out display by control of the first OS at the display device.

(14) When carrying out display by control of the second OS only, there are cases in which constraints arise in the display for the user. However in accordance with the display control device for a vehicle of the fourth aspect, in a case in which a problem no longer is, the display is immediately returned to the display of before the problem arose, and constraints on the display for

a user can thereby be kept to a minimum.

(15) In a display control device for a vehicle of a fifth aspect, in the display control device for a vehicle of any one of the second aspect to the fourth aspect, the at least one processor is configured to acquire information as to whether or not there is a problem with or a possibility of a problem with display of the display device, from an external information processing device.

(16) In the display control device for a vehicle of the fifth aspect, redundancy and accuracy improve because decentralization of the system is possible due to an external information processing device judging whether or not there is a problem or the possibility thereof with displaying by the display device.

(17) A vehicle display device of a sixth aspect includes: the display device; and the display control device for a vehicle of any one of the first aspect to the fifth aspect.

(18) In the vehicle display device of the sixth aspect, display control of a display device installed at a vehicle is carried out by a processor executing a first OS and a second OS. The display device is configured to be able to be seen by a user. In a case in which there is a problem with or the possibility of a problem with displaying by the display device, the at least one processor carries out display on the display device by only control by the second OS. Therefore, in accordance with the present structure, display for the user can be ensured at the vehicle display device that has a display device.

(19) A vehicle of a seventh aspect includes: the display device; and the display control device for a vehicle of any one of the first aspect to the fifth aspect.

(20) In the vehicle of the seventh aspect, display control of a display device installed at a vehicle is carried out by a processor executing a first OS and a second OS. The display device is configured to be able to be seen by a user. In a case in which there is a problem with or the possibility of a problem with displaying by the display device, the at least one processor carries out display on the display device by only control by the second OS. Therefore, in accordance with the present structure, at the vehicle, display for the user can be ensured, and traveling safety can thereby be ensured.

(21) A display control method for a vehicle of an eighth aspect executes, by a processor, processings of: executing a first OS, which carries out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands; and, in a case in which there is a problem with or a possibility of a problem with display of the display device, carrying out display by only control of the second OS at the display device.

(22) In the display control method for a vehicle of the eighth aspect, display control of a display device installed at a vehicle is executed by a processor executing a first OS and a second OS. The display device is configured to be able to be seen by a user. In a case in which there is a problem with or the possibility of a problem with displaying by the display device, the computer executes display on the display device by only control by the second OS. Therefore, in accordance with the present method, display for the user can be ensured in cases in which display control of a display device installed at a vehicle is carried out by plural OSs.

(23) A non-transitory storage medium of a ninth aspect stores a program executable by a processor to perform a process. the process comprising: executing a first OS, which carries out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands; and, in a case in which there is a problem with or a possibility of a problem with display of the display device, carrying out display by only control of the second OS at the display device.

(24) In a program that is stored on the non-transitory storage medium of the ninth aspect, display control of a display device installed at a vehicle is executed by a processor executing a first OS and a second OS. The display device is configured to be able to be seen by a user. In a case in which

there is a problem with or the possibility of a problem with displaying by the display device, the at least one processor has display carried out at the display device by only control by the second OS. Therefore, in accordance with the program that is stored on this non-transitory storage medium, display for the user can be ensured in cases in which display control of a display device installed at a vehicle is carried out by plural OSs.

(25) In accordance with the present disclosure, display for a user can be ensured in cases in which display control of a display device installed at a vehicle is carried out by plural OSs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the present invention will be described in detail based on the following figures, wherein:

(2) FIG. 1 is a schematic structural view of a vehicle that includes a vehicle display device relating to a first embodiment;

(3) FIG. 2 is a block drawing illustrating hardware structures of the vehicle display device relating to the first embodiment;

(4) FIG. 3 is a block drawing illustrating functional structures of an electronic control unit of the vehicle display device relating to the first embodiment;

(5) FIG. 4 is a flowchart explaining main portions of control operations of the electronic control unit of the vehicle display device relating to the first embodiment;

(6) FIG. 5A is an example of a display control screen of the electronic control unit of the vehicle display device relating to a modified example of the first embodiment, and is an example of a display control screen by an RTOS executing section;

(7) FIG. 5B is an example of a display control screen of the electronic control unit of the vehicle display device relating to the modified example of the first embodiment, and is an example of a control screen relating to superposed display by a general-purpose OS executing section and the RTOS executing section; and

(8) FIG. 6 is a flowchart explaining main portions of control operations of the electronic control unit of the vehicle display device relating to the modified example of the first embodiment.

DETAILED DESCRIPTION

(9) Specific embodiments that apply the present disclosure are described in detail hereinafter with reference to the drawings. However, the present disclosure is not limited to the following embodiments. Further, the following description and the drawings are simplified when appropriate for clearer explanation.

(10) In the present disclosure, a processor executes a first OS (Operating System), which carries out display control of a display device installed at the vehicle and viewed by a user of the vehicle, and a second OS that is an OS that carries out display control of the display device and ensures time limits on command execution times. In a case in which there is a problem with or the possibility of a problem with displaying by the display device, the at least one processor carries out display in accordance with the control of the second OS at the display device.

First Embodiment

(11) FIG. 1 is a schematic structural view of a vehicle **1** that includes a display device **10** for a vehicle relating to a first embodiment. As illustrated in FIG. 1, a meter display **102** is set so as to be within the visual field of a user of the vehicle **1**. The meter display **102** is an example of the display device at the display device **10** for a vehicle, and any arbitrary display device that is installed at a vehicle, such as a head-up display or a center display or the like, may be used therefor.

(12) (Hardware Structures of Display Device **10** for Vehicle)

(13) FIG. 2 is a block drawing illustrating hardware structures of the display device **10** for a

vehicle. As illustrated in FIG. 2, the display device **10** for a vehicle includes an electronic control unit (hereinafter called "ECU") **100**, the meter display **102**, a center display **104**, operation switches **112**, vehicle driving devices **106**, onboard sensors **108**, an external communication interface (external communication I/F) **114**, and the like. The ECU **100** is an example of the display control device for a vehicle. At the display device **10** for a vehicle, the ECU **100** is connected to the meter display **102**, the center display **104**, the operation switches **112**, various onboard switches and the like so as to be able to communicate therewith.

(14) Further, the ECU **100** is connected by a CAN (Controller Area Network) bus **116** also to the vehicle driving devices **106**, the onboard sensors **108**, the external communication I/F **114**, other ECUs, onboard equipment and the like so as to be able to communicate therewith.

(15) The ECU **100** of the display device **10** for a vehicle has the function of carrying out display control of the meter display **102**. The ECU **100** is structured to include a CPU (Central Processing Unit) **1000**, a ROM (Read Only Memory) **1001**, a RAM (Random Access Memory) **1002**, an eMMC (embedded Multi Media Card) **1003**, an input/output interface (input/output I/F) **1004** and a communication interface (communication I/F) **1005**. These respective structures are connected so as to be able to communicate with one another through an internal bus **1006**.

(16) The CPU **1000** is a central computing processing unit, and executes various programs and controls the respective sections. Namely, the CPU **1000** reads-out programs from the ROM **1001** and the eMMC **1003**, and executes the programs by using the RAM **1002** as a workspace. The CPU **1000** carries out control of the above-described respective structures, and various types of computing processings, in accordance with the programs stored in the ROM **1001** and the eMMC **1003**. The CPU **1000** may be a multiprocessor, or may be a multicore processor. The CPU **1000** is an example of the processor.

(17) The ROM **1001** stores various programs and various data. The RAM **1002** temporarily stores programs and data as a workspace. The eMMC **1003** is a non-transitory storage medium that stores various programs, including operating systems, and various data. In the present embodiment, an eMMC is used as the storage, but the storage may be structured by a storage device such as an HDD (Hard Disk Drive), an SSD (Solid State Drive), an SSHD (Solid State Hybrid Drive) or the like.

(18) The input/output I/F **1004** is an interface for carrying out input/output of information from and to equipment connected at the exterior, and has a USB port, a DVI port or the like. The meter display **102**, the center display **104**, the operating switches **112**, and various equipment that are installed at the vehicle, such as a touch pad, switches, a microphone, a camera and the like, are connected to the input/output I/F **1004**. The ECU **100** is used also at the time of carrying out display control of the meter display **102**.

(19) The communication I/F **1005** is an interface for carrying out communication with other equipment installed at the vehicle **1**. The vehicle driving devices **106**, the onboard sensors **108**, the external communication I/F **114**, and other ECUs and onboard equipment and the like are connected to the communication I/F **1005** so as to be able to communicate therewith, by the CAN bus **116** that is a bus of the CAN standard. The communication standards may be changed appropriately, such as MOST, IDB-1394 or an onboard Ethernet® may be used in the transfer of large amounts of data such as digital content or the like, and a Local Interconnect Network (LIN) may be used in a system that does not require that high of a transfer speed, and the Safe-by-Wire standard that is low speed but highly reliable may be used in occupant protecting devices such as the airbags and seatbelts and the like, and the like.

(20) The meter display **102** is a display device that is set at a position within the visual field of the user of the vehicle **1** who is in a driving posture. Due to display control being carried out by the ECU **100** through the input/output I/F **1004**, the meter display **102** displays information for the user of the vehicle **1**, and carries out vehicle operation assist. Further, the ECU **100** is connected to various meters that are installed at the vehicle. Therefore, information relating to various vehicle

functions such as, the first and foremost, the speed of the vehicle **1**, the number of revolutions of the engine, the remaining fuel amount, the states of the gears and the battery, the display lights and the warning lights, and the like, are displayed on the meter display **102**.

(21) The center display **104** is a display device that is mounted in a vicinity of the vehicle transverse direction center of the instrument panel of the vehicle **1**. In the same way as the meter display **102**, due to display control being carried out by the ECU **100** through the input/output I/F **1004**, the center display **104** displays information for the user of the vehicle **1**, and carries out vehicle operation assist. The center display **104** has the function of displaying map information that guides the user to the destination, and news and weather information, and playing back music, and the like on the basis of signals inputted by the user of the vehicle **1**. Further, the display control section of the center display **104** and the display control section of the meter display **102** may have information in common, or may coordinate information with one another by one of these display control sections providing, to the other, information that is the basis for preparing displayed images, such as map information of the periphery and the traveling state of the vehicle and the like, that are displayed on the respective displays.

(22) The operation switches **112** are set at positions at which they can be operated by the user of the vehicle, and are used at the time when the user carries out changing of the displays of the meter display **102** and the center display **104**, control of the audio system, and vehicle operations such as implementing/stopping the drive assist system and the like.

(23) The vehicle driving devices **106** are devices for driving that are provided at the vehicle **1**, and drive the vehicle **1** on the basis of signals inputted by the user. Namely, the vehicle driving devices **106** include motors, valves and actuators that are mounted to respective portions of the body and respective portions of the engine. These devices may be connected to the CAN bus **116** via ECUs used exclusively therefor.

(24) The onboard sensors **108** are sensors provided at the vehicle **1**, and are provided for the purpose of acquiring various information for the vehicle **1** and driving assist therefor. The onboard sensors **108** include sensors that are mounted to respective portions of the body and respective portions of the engine and the like of the vehicle **1**. These sensors may be connected to the CAN bus **116** via ECUs used exclusively therefor.

(25) The external communication I/F **114** is provided in order to carry out communication with external devices such as external centers or the like. Such communication includes the transmitting of information acquired from other onboard equipment through the CAN bus **116**, and the receiving of information of the periphery of the vehicle and various service information such as maintenance information and the like, from external devices. The external communication I/F **114** has a communication module, and may carry out communication by wired or wireless communication using any of various standards such as, for example, CAN, Ethernet®, FDDI (Fiber Distributed Data Interface), Wi-Fi®, I2C (Inter-Integrated Circuit), Zigbee®, Bluetooth®, LTE (Long Term Evolution), and the like. Further, an arbitrary control unit may have the function of carrying out communications with the exterior. For example, the ECU **100** may be configured to also have the function of carrying out communications with the exterior.

(26) (Functional Structures of Display Device for Vehicle Relating to Present Embodiment)

(27) FIG. **3** is a block drawing illustrating functional structures of the vehicle display device relating to the first embodiment.

(28) In the display device **10** for a vehicle that is provided at the vehicle **1**, the ECU **100** carries out display control of the meter display **102** on the basis of information acquired from onboard equipment such as the vehicle driving devices **106**, the onboard sensors **108**, the external communication I/F **114**, the center display **104**, the operation switches **112** and the like.

(29) A hypervisor **1007** functions due to a program that is stored in the ROM **1001** or the eMMC **1003** that is executed by the CPU **1000** by using the RAM **1002** as a workspace, at the time when the power is turned on. The hypervisor **1007** creates virtual machines on the hardware, and can

operate the OSs. Specifically, hardware such as the CPU **1000**, the ROM **1001**, the RAM **1002**, the eMMC **1003**, the input/output I/F **1004**, the communication I/F **1005** and the like are controlled on the basis of control that is intended by a guest OS operated at the hypervisor **1007**. Further, the hypervisor **1007** always monitors the display of the meter display **102**. The hypervisor **1007** is an example of the system that is used at the time when the CPU **1000** switches the OS that carries out display control of the meter display **102**.

(30) In the present embodiment, as an example of the first OS that is used as the guest OS, a general-purpose OS is used for operation of a display control application by which the ECU **100** carries out display control of the meter display **102**. Further, an RTOS that is an OS that ensures the time limits of execution times of commands is used as an example of the second OS. Examples of the general-purpose OS are AGL (Automotive Grade Linux)®, AAOS (Android Automatic Operating System)® and the like. Examples of the RTOS are QNX®, µITRON®, RTLinux®, and the like. Further, in order to ensure the time limit of the execution time of a command, an OS is selected that fulfills requests from the user application, such as having a scheduling rule that ensures that a high-priority task, which fulfills a request for the worst case value until the processing routine is called, is reliably executed after the occurrence of a hardware interrupt that fulfills a request for the worst case value needed in a case in which a service call at which a wait does not arise is called, or the like.

(31) The general-purpose OS is executed by a general-purpose OS executing section **1008**, and operates the display control application of the meter display **102** on the system of the general-purpose OS that is executed. Further, similarly, the RTOS is executed by an RTOS executing section **1009**, and operates the display control application of the meter display **102** on the system of the RTOS that is executed. In the display control of the meter display **102**, both the display control application of the meter display **102** that is executed by the general-purpose OS executing section **1008** and the display control application of the meter display **102** that is executed by the RTOS executing section **1009** are always operating, and display control in accordance with the display control applications on the respective OSs is carried out by the hypervisor **1007** appropriately switching the OS on which the application that carries out the display control operates. Further, although the present embodiment is structured such that both applications are always operating, there may be a structure in which the applications are stopped temporarily, and the applications are executed again when needed.

(32) Further, in the present embodiment, display control of the center display **104** is carried out similarly by the same hypervisor **1007**, but may be carried out on different hardware. Further, the display control section may carry out communication, and, when information is shared, may use the information as display information.

(33) (Flowchart Describing Main Portions of Control Operations)

(34) FIG. **4** is a flowchart describing main portions of the control operations of the ECU **100** of the vehicle **1** relating to the first embodiment. The following processings are realized due to the CPU **1000** at the ECU **100** functioning as the hypervisor **1007**, the general-purpose OS executing section **1008** and the RTOS executing section **1009**.

(35) The flowchart of FIG. **4** is executed each time the ignition of the vehicle **1** is turned on. First, in step **S100** (the word “step” will be omitted hereinafter), the CPU **1000** detects that the ignition of the vehicle **1** has been turned on.

(36) In **S102**, the CPU **1000** executes the hypervisor **1007**. Specifically, when the ignition is turned on in **S100** that is the previous step, and power is automatically supplied to the ECU **100** as well, the ECU **100** executes the hypervisor **1007** due to a control program stored in the eMMC **1003** that is executed by the CPU **1000** by using the RAM **1002** as a workspace.

(37) In **S104**, the CPU **1000** executes the general-purpose OS and the RTOS. Specifically, the general-purpose OS and the RTOS are executed on the hypervisor **1007** that was executed in **S102**, and preparations for display control of the meter display **102** are carried out.

- (38) In **S106**, the CPU **1000** carries out display control of the meter display **102** by the RTOS. Specifically, when the RTOS is executed, and preparations for display control of the meter display **102** by the display control application on the RTOS are completed, display control of the meter display **102** by the RTOS is carried out.
- (39) In **S108**, the CPU **1000** judges whether or not preparations for display control by the general-purpose OS have been completed. Specifically, the CPU **1000** judges whether preparations for display control of the meter display **102** by the display control application that operates on the executed general-purpose OS have been completed. If the CPU **1000** judges that preparations for display control by the general-purpose OS have been completed (in the case of Yes in **S108**), the CPU **1000** advances to **S110**. On the other hand, if the CPU **1000** judges that preparations for display control by the general-purpose OS have not been completed (in the case of No in **S108**), the CPU **1000** returns to **S106**.
- (40) In **S110**, the CPU **1000** switches the OS that is carrying out display control to display control by the general-purpose OS. Namely, when preparations for display control by the general-purpose OS have been completed, the hypervisor **1007** switches the OS that carries out display control from display control by the RTOS to display control by the general-purpose OS.
- (41) In **S112**, the CPU **1000** judges whether or not there is a problem, or the possibility of a problem, with the display of the meter display **102**. If the CPU **1000** judges that there are no problems and no possibility thereof in the display of the meter display **102**, i.e., if there is neither a problem nor the possibility of a problem (in the case of No in **S112**), the CPU **1000** advances to **S114**. On the other hand, if the CPU **1000** judges that there is a problem or the possibility thereof in the display of the meter display **102**, i.e., if there is either a problem or the possibility of a problem (in the case of Yes in **S112**), the CPU **1000** advances to **S116**.
- (42) In **S114**, the CPU **1000** executes display control by the general-purpose OS. Namely, in a case in which display control by the general-purpose OS has been executed up until then, the display control by the general-purpose OS is continued. In a case in which display control by the RTOS has been executed up until then, the display control is switched to display control by the general-purpose OS.
- (43) In **S116**, the CPU **1000** executes display control by the RTOS. Namely, in a case in which display control by the general-purpose OS has been executed up until then, the display control is switched to display control by the RTOS. In a case in which display control by the RTOS has been executed up until then, the display control by the RTOS is continued.
- (44) In **S118**, the CPU **1000** judges whether or not a given time period has elapsed. If the CPU **1000** judges that the given time period has elapsed (in the case of Yes in **S118**), the CPU **1000** returns to **S112**. On the other hand, if the CPU **1000** judges that the given time period has not elapsed (in the case of No in **S118**), the CPU **1000** repeats **S118**. In accordance with the present structure, if there is no problem and no possibility thereof, it is possible to return to display control by the original, general-purpose OS.
- (45) As described above, in accordance with the present embodiment, in a case in which, after display control of the meter display **102** by the general-purpose OS executing section **1008**, there is a problem with the display of the meter display **102** or the possibility of a problem, display control is switched to display control by the RTOS executing section **1009**. Due thereto, display for the user of the display device can be ensured even in cases in which there is a problem with, or there is the possibility of a problem with, the display of the meter display **102**. In particular, in a case in which the meter display **102** is used as the display device, display that is directed toward the driver who is the user can be ensured. In accordance with the present embodiment, at the time when a driver drives the vehicle **1**, display for the driver is ensured, and therefore, the present embodiment can contribute to the traveling safety of the vehicle **1**.

Modified Example

- (46) In the first embodiment, at the time of controlling the display of the meter display **102**, there is

display control by the general-purpose OS or display control by the RTOS. However, superposed display control by both the general-purpose OS and the RTOS may be carried out.

(47) FIG. 5A is an example of a display control screen in accordance with the RTOS executing section **1009** of the electronic control unit **100** of the display device **10** for a vehicle relating to a modified example of the first embodiment. In the display control by the RTOS executing section **1009**, the speed of preparing for display control is prioritized over the amount of information, and therefore, in order to carry out driving assist, the amount of information that is to be displayed on the meter display **102** is limited, and display of only some important display information **1020** is carried out. An example of the important display that is displayed by the important display information **1020** is information such as telltales expressing problems with the ABS or problems with the engine, the hazard lights, information of the current shift position, the cumulative distance travelled, the current speed, the remaining amount of gasoline, the water temperature and the like.

(48) FIG. 5B is an example of a superposed display control screen in accordance with the general-purpose OS executing section **1008** and the RTOS executing section **1009** of the electronic control unit **100** of the display device **10** for a vehicle relating to the modified example of the first embodiment. In the superposed display control by the general-purpose OS executing section **1008** and the RTOS executing section **1009**, display control by the general-purpose OS executing section **1008** is superposed on the display control of the RTOS executing section **1009** of FIG. 5A. The general-purpose OS executing section prioritizes high-level image display such as the amount of information and rendering and the like, over the speed of preparation of the display control, and therefore, display control of various information is possible, and display of generally displayed information **1021** can be carried out. Examples of the general display that is displayed by the generally displayed information **1021** include information of the tachometer, behavior predicting information of the drive assist system in 3D display, information such as the time and the like, and decorative display.

(49) FIG. 6 is a flowchart explaining main portions of control operations of the electronic control unit of the vehicle display device relating to the modified example of the first embodiment.

(50) Processings from **S200** to **S208** are similar to the processings of **S100** to **S108** of the above-described first embodiment. In **S110** of the above-described first embodiment that corresponds to **S210** that is the step that follows next, display control of the meter display **102** is carried out by only the general-purpose OS. However, in **S210** of the present modified example, superposed display control, in which display control by the RTOS is added to the display control by the general-purpose OS, is carried out. Due thereto, in both the RTOS and the general-purpose OS, there is no need to carry out control for displaying similar information, and an improvement in efficiency at the time of manufacturing the manufactured product, and a reduction in the memory capacity for information, and the like are possible.

(51) The processing of next **S212** is similar to the processing of **S112**. Further, **S214** is similar to above-described **S210**. The processings of **S216**, **S218** are similar to **S116**, **S118**.

(52) Further, although the first embodiment is structured such that the hypervisor **1007** judges whether there is a problem or the possibility thereof with the display of the meter display **102**, there may be a structure in which an information processing device, which judges whether there is a problem or the possibility thereof with the display of the meter display **102**, is provided separately at the exterior, and information is acquired. In accordance with this structure, decentralization of the system is possible due to the external information processing device judging whether or not there is a problem or the possibility thereof with display of the display device, and the redundancy and accuracy improve.

(53) Further, the first embodiment is structured such that the hypervisor **1007** judges whether there is a problem or the possibility thereof with the display of the meter display **102** in **S112** each time a given time period elapses. However, there may be a structure in which, in a case of acquiring fail information expressing that display control of the meter display **102** by the general-purpose OS

executing section **1008** could not be carried out normally, the step of **S116** is executed immediately without waiting out the given time period.

(54) In the first embodiment, the structure in which the general-purpose OS executing section **1008** and the RTOS executing section **1009** further operate on the hypervisor **1007** that operates on hardware is an example of the system that is used at the time when the CPU **1000** switches the OS that carries out display control of the meter display **102**. However, an arbitrary, known structure may be used, such as hardware being controlled indirectly through another host OS that is between the hardware and the hypervisor **1007**, or the like.

(55) In the first embodiment, there is a structure in which the general-purpose OS executing section **1008** and the RTOS executing section **1009** operate on the hypervisor **1007**. However, the number of host OSs is not limited to two, and there may be a structure in which three or more host OSs operate in order to provide even more redundancy. Further, there may be a structure in which the superposed display of the modified example also is superposed display by three or more host OSs.

(56) Although the present disclosure has been described on the basis of the respective drawings and embodiments, a person skilled in the art can easily make various modifications and changes on the basis of the present disclosure. The present disclosure is not limited to the above-described embodiments, and such modifications and changes are to be considered as falling within the scope of the present disclosure. For example, functions and the like that are included in the respective means and the respective steps or the like can be rearranged provided that they are not logically contradictory. Plural means or steps or the like may be combined into a single means or step, or may be divided so as to be carried out by plural units and/or arbitrary servers. Further, devices and methods that carry out operations relating to the present embodiments at the control processing section of the display control device for a vehicle, and programs that cause operations relating to the present embodiment to be carried out, also are included within the scope of the present disclosure.

(57) Further, any of various types of processors other than a CPU may execute the respective processings that are executed due to the CPU **1000** reading-in software (programs) in the above-described embodiments. Examples of processors in this case include PLDs (Programmable Logic Devices) whose circuit structure can be changed after production such as FPGAs (Field-Programmable Gate Arrays) and the like, and dedicated electrical circuits that are processors having circuit structures that are designed for the sole purpose of executing specific processings such as ASICs (Application Specific Integrated Circuits) and the like, and the like. Further, the above-described respective processings may be executed by one of these various types of processors, or may be executed by a combination of two or more of the same type or different types of processors (e.g., plural FPGAs, or a combination of a CPU and an FPGA, or the like). Further, the hardware structures of these various types of processors are, more specifically, electrical circuits that combine circuit elements such as semiconductor elements and the like.

(58) The above embodiments describe a form in which the respective programs are stored in advance (are installed) on a non-transitory, computer-readable storage medium. For example, the program for control at the ECU **100** is stored in advance in the eMMC **1003**. However, the present disclosure is not limited to this, and the respective programs may be provided in forms of being stored on a non-transitory storage medium such as a CD-ROM (Compact Disc Read Only Memory), a DVD-ROM (Digital Versatile Disc Read Only Memory), a USB (Universal Serial Bus) memory, or the like. Further, the programs may be in forms of being downloaded from an external device over a network.

Claims

1. A display control device for a vehicle, comprising: a memory; and at least one processor coupled to the memory, wherein the at least one processor is configured to: execute a first OS, which carries

out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands, and in a case in which there is a problem with or a possibility of a problem with display of the display device, carry out display by only control of the second OS at the display device.

2. The display control device for a vehicle of claim 1, wherein the at least one processor is configured to: be able to switch between display by control of the first OS and display by control of the second OS, and in the case in which there is a problem with or a possibility of a problem with display of the display device, carry out display by control of the second OS instead of display by control of the first OS at the display device.

3. The display control device for a vehicle of claim 1, wherein the at least one processor is configured to: be able to carry out superposed display of display by control of the first OS and display by control of the second OS at the display device, and in the case in which there is a problem with or a possibility of a problem with display of the display device, stop display by control of the first OS at the display device.

4. The display control device for a vehicle of claim 1, wherein the at least one processor is configured to, in a case in which the problem disappears, carry out display by control of the first OS at the display device.

5. The display control device for a vehicle of claim 1, wherein the at least one processor is configured to acquire information as to whether or not there is a problem with or a possibility of a problem with display of the display device, from an external information processing device.

6. A vehicle display device, comprising: the display device; and the display control device for a vehicle of claim 1.

7. A vehicle comprising: the display device; and the display control device for a vehicle of claim 1.

8. A display control method for a vehicle that is executed by a processor, the display control method comprising: executing a first OS, which carries out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands; and in a case in which there is a problem with or a possibility of a problem with display of the display device, carrying out display by only control of the second OS at the display device.

9. A non-transitory storage medium storing a program executable by a processor to perform a process, the process comprising: executing a first OS, which carries out display control of a display device installed at a vehicle and viewed by a user of the vehicle, and a second OS that carries out display control of the display device and ensures time limits of execution times of commands; and in a case in which there is a problem with or a possibility of a problem with display of the display device, carrying out display by only control of the second OS at the display device.
